Practical-6

AIM: Configure and evaluate integrated DNS and FTP services across an interconnected LAN–WAN environment, implement role-based user access and file-transfer permissions, analyze DNS resolution and FTP communication using Packet Tracer Simulation Mode, troubleshoot service-related faults, and compare FTP with secure alternatives such as FTPS and SFTP.

Tools required: Cisco Packet Tracer

Submission: After writing an answer into this word document, Student needs to change name to his ID followed by practical number. Ex 25CS005_Pr6.docx. Upload on assignment segment.

Rubrics: Nicely drafted document with clarity in answers leads to full marks. Otherwise, submission carries proportional marks.

Recommended to type, avoid copy-paste to increase your typing skill.

Note: While applying IP address, student needs to allocate IP address as per his/her student ID. For Example, if student ID is 25CS005 then IP address allocation for the first network should start with 5.0.0.0. For subsequent networks, it should start with ID+1 i.e. 6.0.0.0, 7.0.0.0 and so on.

Student should be able to

1. Deploy DNS and FTP services across an already-routed LAN–WAN topology and verify application-layer reachability.
2. Configure DNS resource records and compare FTP access using an IP address and a domain name.
3. Configure an FTP server with administrator, faculty and student accounts having appropriate read, write and delete permissions.
4. Perform and evaluate FTP authentication, file listing, upload, download, deletion and access-denied operations.
5. Analyze DNS query–response exchanges, TCP connection establishment, FTP control and data communication, and relevant port numbers using Simulation Mode.
6. Diagnose DNS, FTP service, authentication and permission faults, and recommend FTP, FTPS or SFTP for different security requirements.

A university has a Main Campus and a Remote Campus connected through a routed WAN.

The Main Campus contains:

- Central DNS server
- Central FTP server
- Administration PC
- Faculty PC
- Main-campus router and switch

The Remote Campus contains:

- Faculty PC
- Student PC
- Remote-campus router and switch

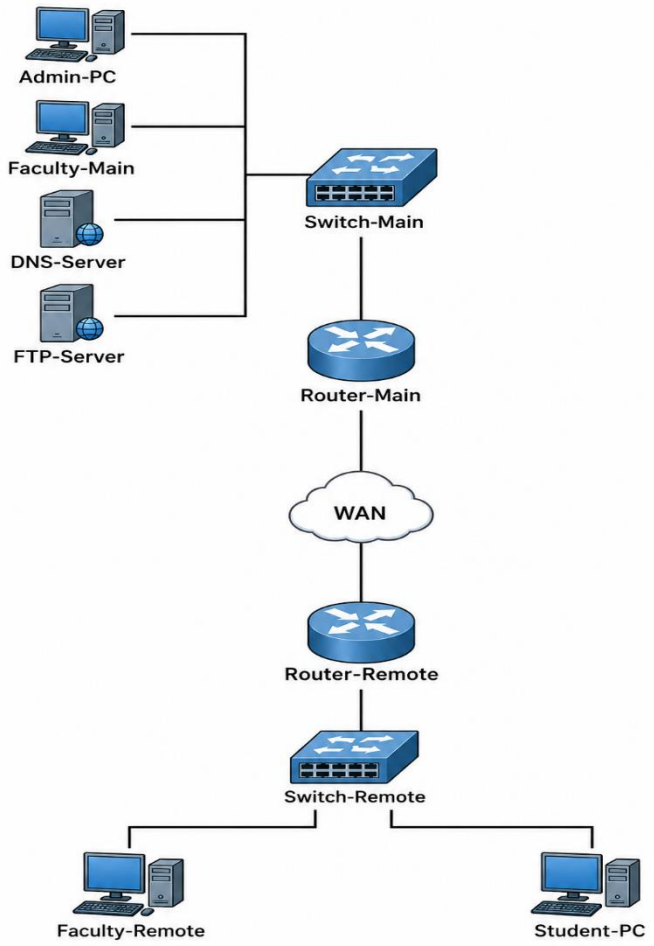
The university requires users to access the FTP server using the domain name:

ftp.university.example

Three categories of users must be configured:

User category	Required access
Administrator	List, read, upload, modify and delete
Faculty	List, read and upload
Student	List and download only

The network must support FTP access both within the Main Campus and from the Remote Campus.
Suggested Topology:



IP-Addressing Plan:

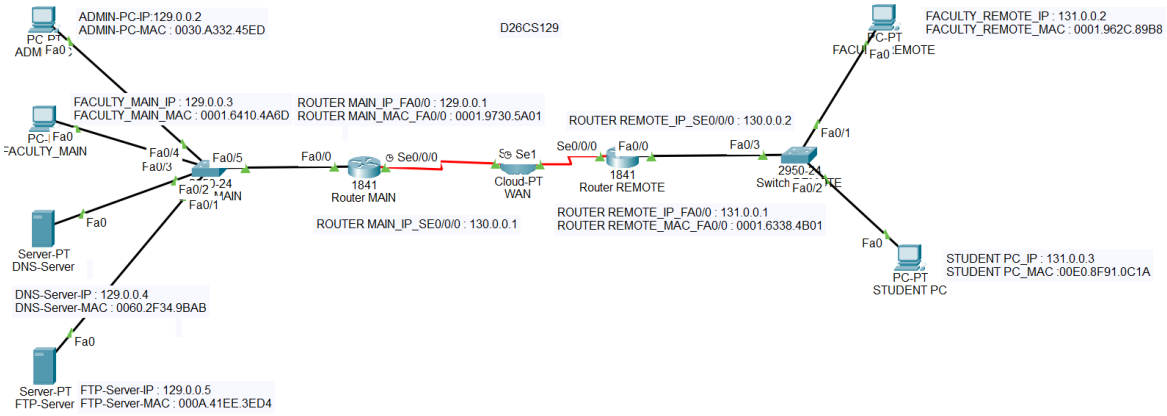
Kindly complete the following table as per your enrolment number and use it for IP assignment strategy using RIPv2 Routing:

Device	IP address	Subnet mask	Default gateway	DNS server
Router-Main LAN	129.0.0.1	255.255.0.0	129.0.0.1	129.0.0.4
DNS Server	129.0.0.4	255.255.0.0	129.0.0.1	129.0.0.4
FTP Server	129.0.0.5	255.255.0.0	129.0.0.1	129.0.0.4
Admin-PC	129.0.0.2	255.255.0.0	129.0.0.1	129.0.0.4
Faculty-Main	129.0.0.3	255.255.0.0	129.0.0.1	129.0.0.4
Router-Main WAN	130.0.0.1	255.255.0.0	-	-
Router-Remote WAN	130.0.0.2	255.255.0.0	-	-
Router-Remote LAN	131.0.0.1	255.255.0.0	-	-
Faculty-Remote	131.0.0.2	255.255.0.0	131.0.0.1	129.0.0.4
Student-PC	131.0.0.3	255.255.0.0	131.0.0.1	129.0.0.4

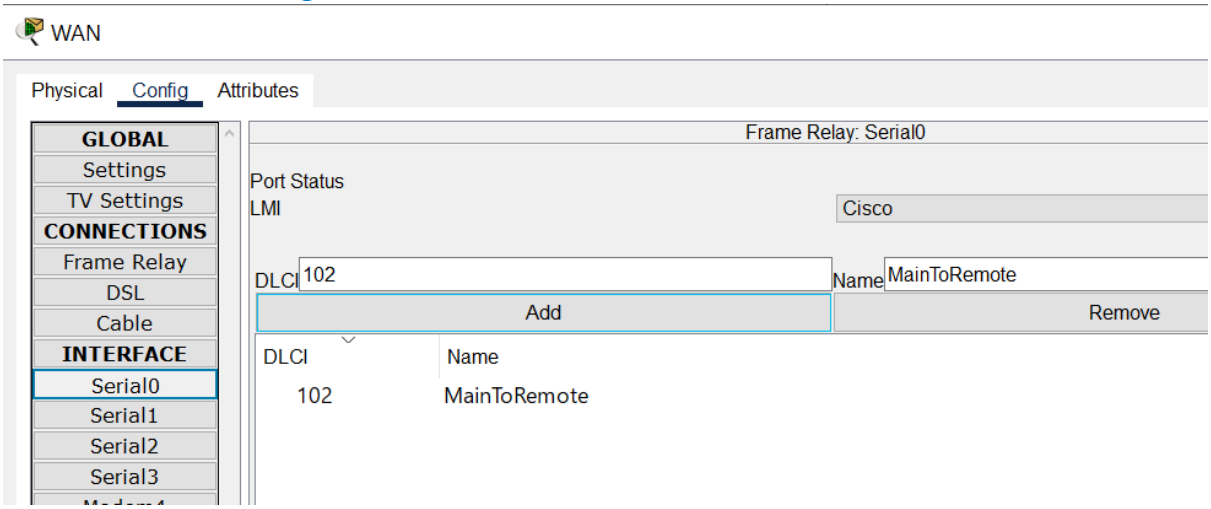
Part A: Basic Network Verification

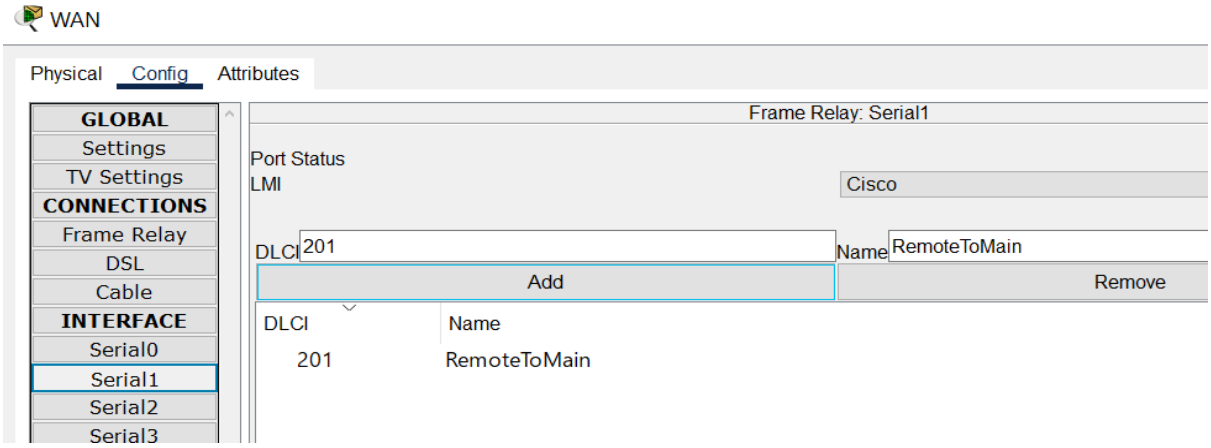
Before configuring DNS and FTP services:

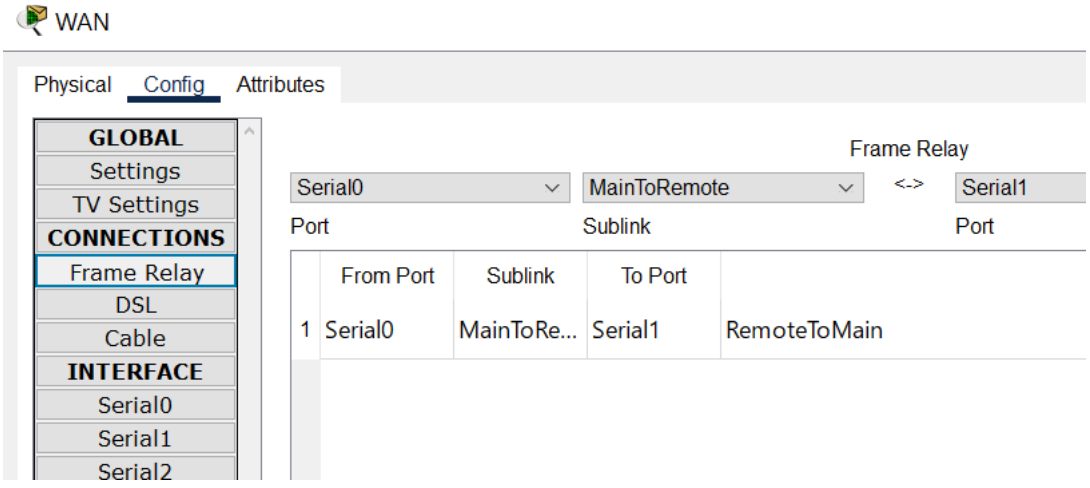
- Create the Main Campus and Remote Campus topology.
- Configure IP addresses and subnet masks.
- Configure default gateways on all end devices.
- Configure routing between the two campuses.



Cloud PT_WAN Configuration:







Another way for ROUTER MAIN-WAN Configuration :

Router-Main> enable
Router-Main# configure terminal
Router-Main(config)# interface Serial0/0/0
Router-Main(config-if)# ip address 125.0.0.1 255.0.0.0

```
Router-Main(config-if)# encapsulation frame-relay
Router-Main(config-if)# frame-relay interface-dlci 102
Router-Main(config-if)# no shutdown
Router-Main(config-if)# exit
```

Another way for ROUTER REMOTE-WAN Configuration :

```
Router-Remote> enable
Router-Remote# configure terminal
Router-Remote(config)# interface Serial0/0/0
Router-Remote(config-if)# ip address 125.0.0.2 255.0.0.0
Router-Remote(config-if)# encapsulation frame-relay
Router-Remote(config-if)# frame-relay interface-dlci 201
Router-Remote(config-if)# no shutdown
Router-Remote(config-if)# exit
```

- Verify interface status.

ROUTER MAIN:

```
Router>enable
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        129.0.0.1       YES manual up                    up
FastEthernet0/1        unassigned      YES unset  down                  down
Serial0/0/0            130.0.0.1       YES manual up                    down
Serial0/0/1            unassigned      YES unset  down                  down
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

ROUTER REMOTE:

```
Router>enable
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        131.0.0.1       YES manual up                    up
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/0/0            130.0.0.2       YES manual up                    down
Serial0/0/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

- Ping the DNS and FTP servers from every client.

ADMIN-PC

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 129.0.0.4

Pinging 129.0.0.4 with 32 bytes of data:

Reply from 129.0.0.4: bytes=32 time=1ms TTL=128
Reply from 129.0.0.4: bytes=32 time=1ms TTL=128
Reply from 129.0.0.4: bytes=32 time<1ms TTL=128
Reply from 129.0.0.4: bytes=32 time<1ms TTL=128

Ping statistics for 129.0.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
C:\>ping 129.0.0.5

Pinging 129.0.0.5 with 32 bytes of data:

Reply from 129.0.0.5: bytes=32 time<1ms TTL=128
Reply from 129.0.0.5: bytes=32 time=1ms TTL=128
Reply from 129.0.0.5: bytes=32 time<1ms TTL=128
Reply from 129.0.0.5: bytes=32 time<1ms TTL=128

Ping statistics for 129.0.0.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

FACULTY_MAIN

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 129.0.0.4

Pinging 129.0.0.4 with 32 bytes of data:

Reply from 129.0.0.4: bytes=32 time<1ms TTL=128
Reply from 129.0.0.4: bytes=32 time<1ms TTL=128
Reply from 129.0.0.4: bytes=32 time<1ms TTL=128
Reply from 129.0.0.4: bytes=32 time<1ms TTL=128

Ping statistics for 129.0.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

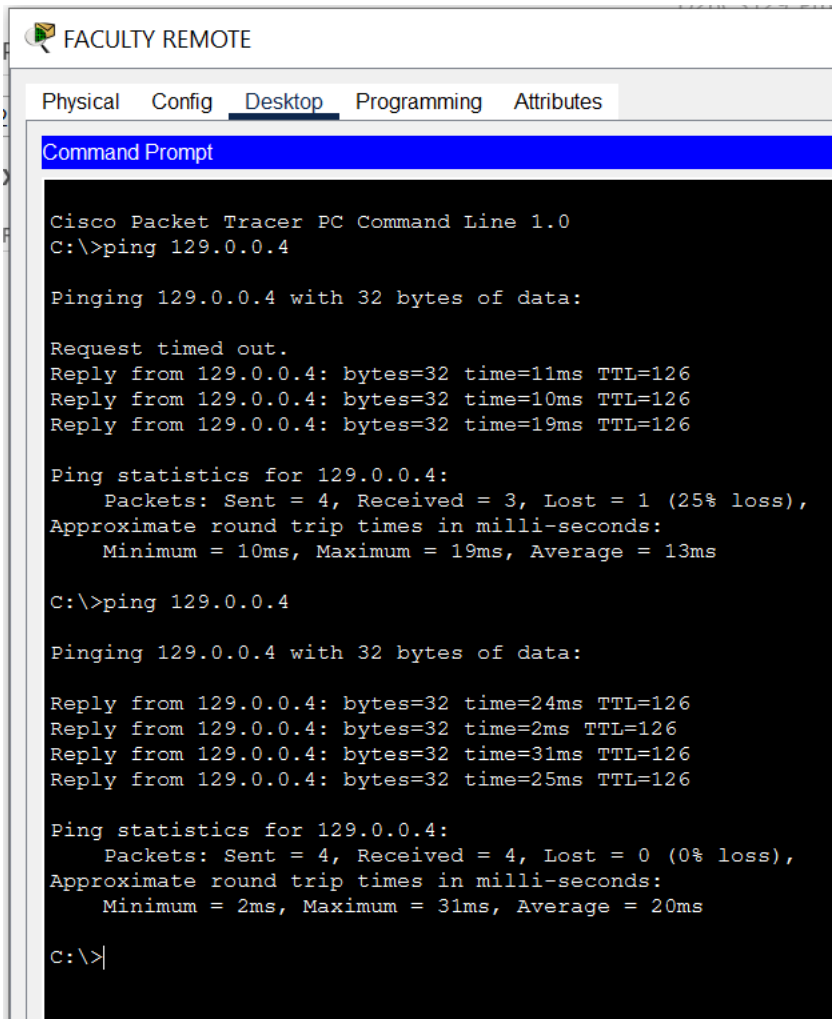
C:\>ping 129.0.0.5

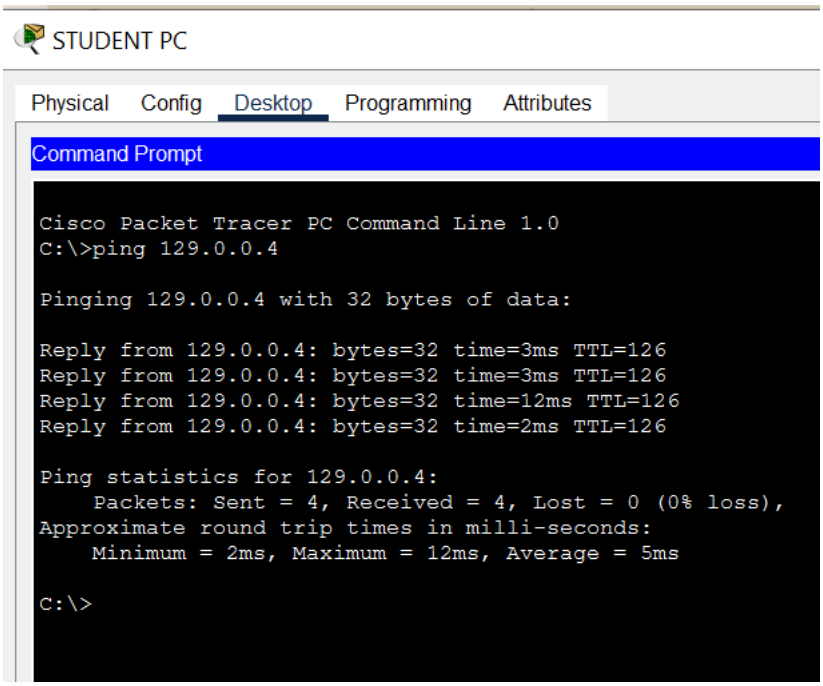
Pinging 129.0.0.5 with 32 bytes of data:

Reply from 129.0.0.5: bytes=32 time<1ms TTL=128
Reply from 129.0.0.5: bytes=32 time<1ms TTL=128
Reply from 129.0.0.5: bytes=32 time=1ms TTL=128
Reply from 129.0.0.5: bytes=32 time=1ms TTL=128

Ping statistics for 129.0.0.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```


FACULTY REMOTE
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 129.0.0.4
Pinging 129.0.0.4 with 32 bytes of data:
Request timed out.
Reply from 129.0.0.4: bytes=32 time=11ms TTL=126
Reply from 129.0.0.4: bytes=32 time=10ms TTL=126
Reply from 129.0.0.4: bytes=32 time=19ms TTL=126
Ping statistics for 129.0.0.4:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 10ms, Maximum = 19ms, Average = 13ms
C:\>ping 129.0.0.4
Pinging 129.0.0.4 with 32 bytes of data:
Reply from 129.0.0.4: bytes=32 time=24ms TTL=126
Reply from 129.0.0.4: bytes=32 time=2ms TTL=126
Reply from 129.0.0.4: bytes=32 time=31ms TTL=126
Reply from 129.0.0.4: bytes=32 time=25ms TTL=126
Ping statistics for 129.0.0.4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 31ms, Average = 20ms
C:\>


STUDENT PC
Physical
Config
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Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 129.0.0.4
Pinging 129.0.0.4 with 32 bytes of data:
Reply from 129.0.0.4: bytes=32 time=3ms TTL=126
Reply from 129.0.0.4: bytes=32 time=3ms TTL=126
Reply from 129.0.0.4: bytes=32 time=12ms TTL=126
Reply from 129.0.0.4: bytes=32 time=2ms TTL=126
Ping statistics for 129.0.0.4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 12ms, Average = 5ms
C:\>

- Verify that the remote-campus clients can reach the FTP server using its IP address.
  STUDENT PC:

```
C:\>tracert 129.0.0.5

Tracing route to 129.0.0.5 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      131.0.0.1
  2    2 ms      3 ms      1 ms      130.0.0.1
  3    *         2 ms      17 ms     129.0.0.5

Trace complete.

C:\>
```

FACULTY PC:

```
C:\>tracert 129.0.0.5

Tracing route to 129.0.0.5 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      131.0.0.1
  2    1 ms      2 ms      3 ms      130.0.0.1
  3    0 ms      12 ms     1 ms      129.0.0.5

Trace complete.
```

- Useful commands: ipconfig, ping <FTP-server-IP>, ping <DNS-server-IP>, tracert <FTP-server-IP>

Connectivity Verification Table:

Source device	Destination	Test used	Result	Observation
Admin-PC	FTP Server	Ping	Successful	FTP server reachable
Faculty-Main	FTP Server	Ping	Successful	FTP server reachable
Faculty-Remote	FTP Server	Ping	Successful	WAN routing working
Student-PC	FTP Server	Ping	Successful	Remote LAN can reach Main LAN

Do not proceed to application-service troubleshooting until IP connectivity is working.

Part B: DNS Service Configuration

On the DNS server:

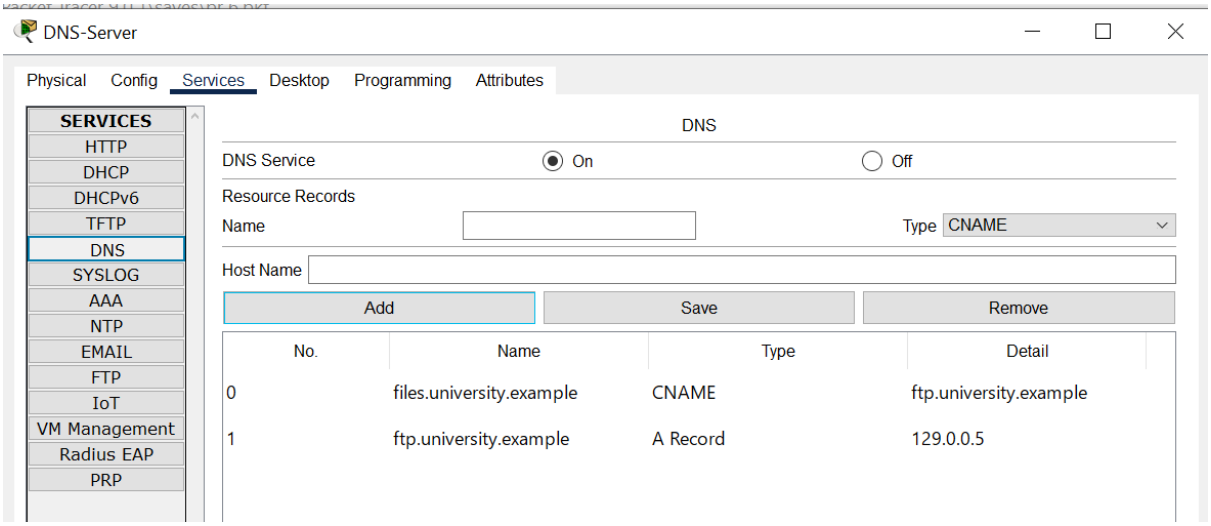
- Open Services → DNS.
- Enable the DNS service.
- Create the following record:

Record name	Record type	Address
`ftp.university.example`	A	FTP-server IP address

An optional alias may also be created:

Alias	Target
`files.university.example`	`ftp.university.example`

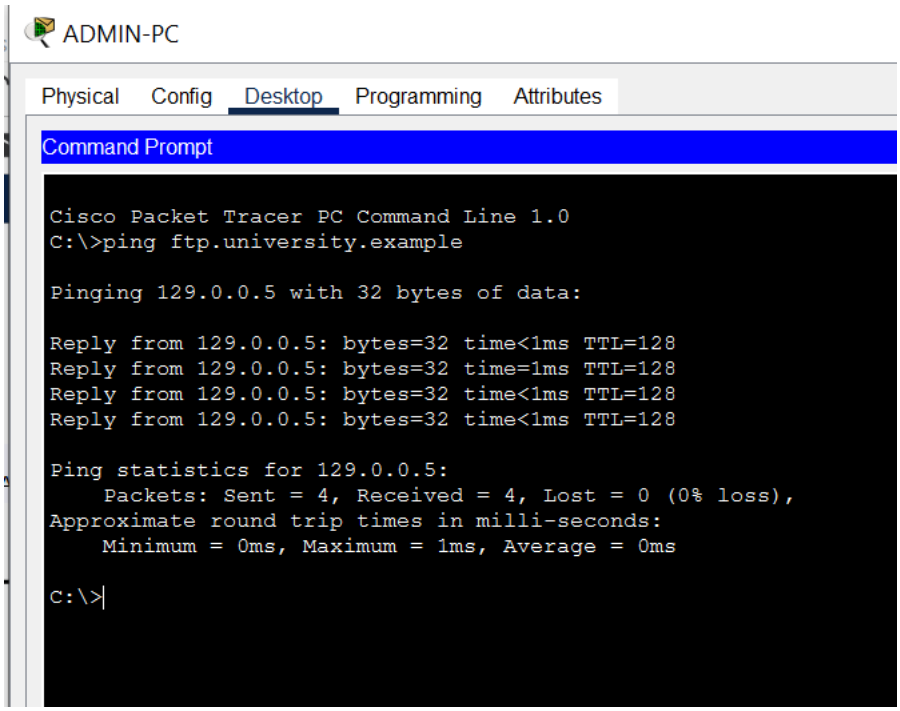
- Configure every client PC with the IP address of the DNS server.

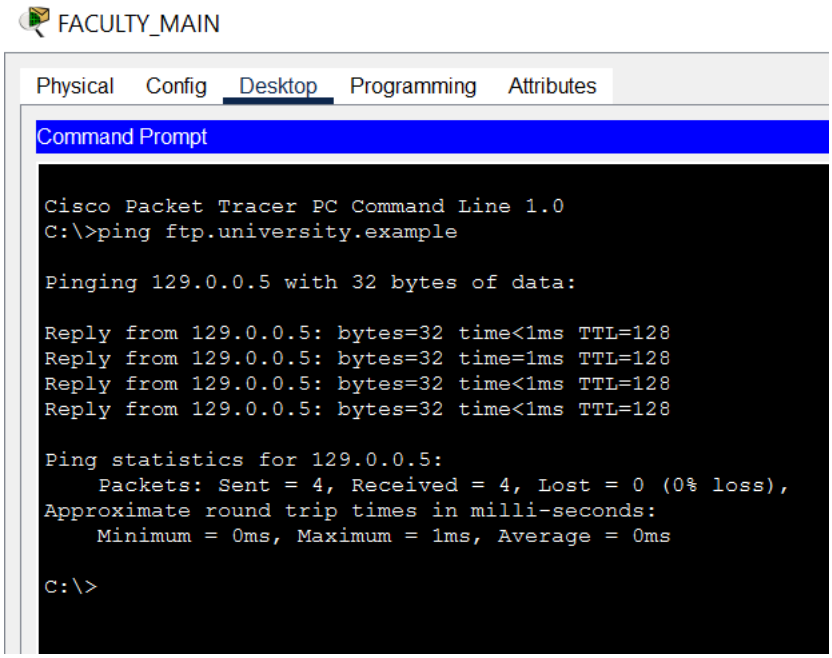


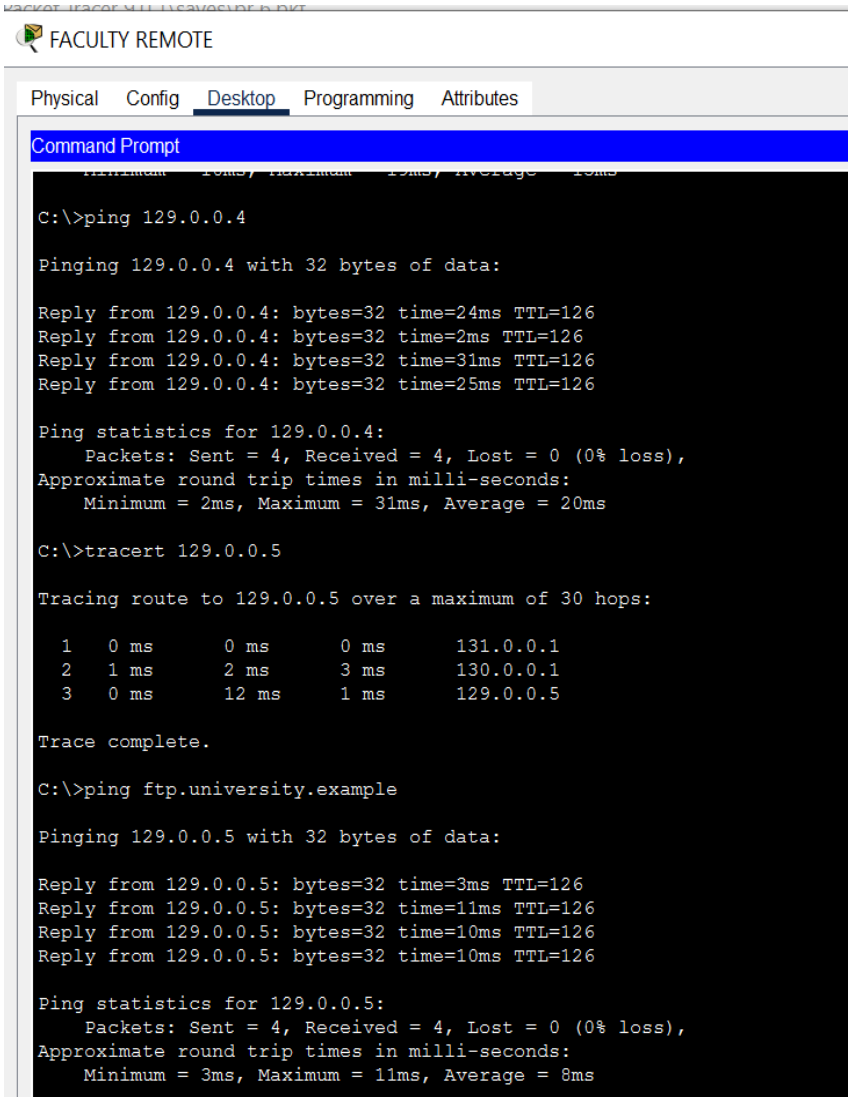
DNS Verification:

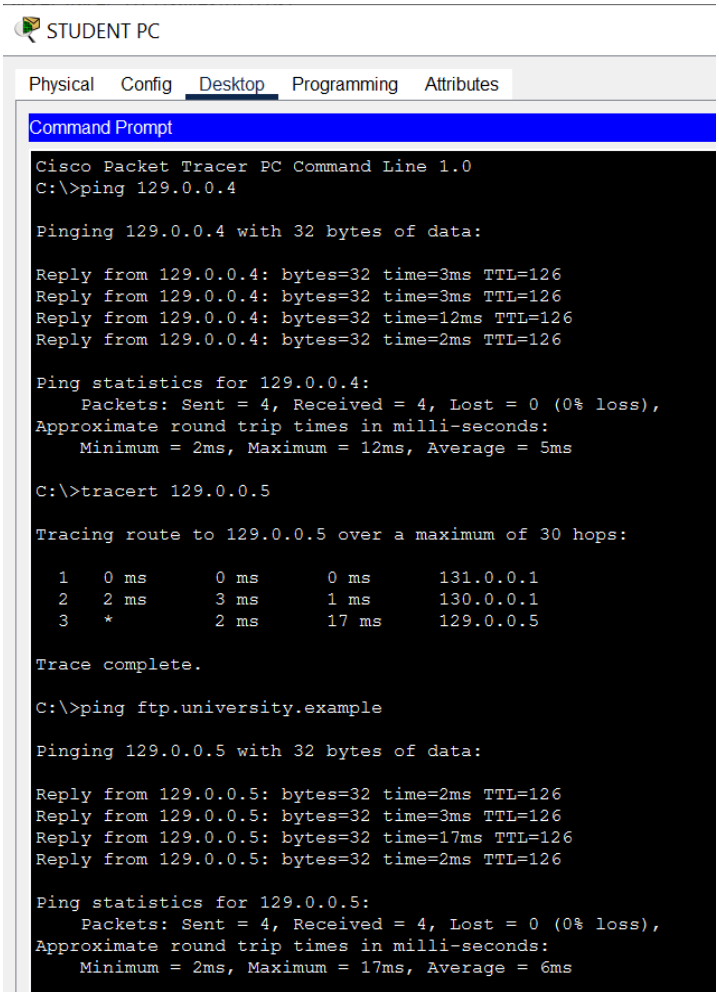
- Ping the FTP server using its IP address.
- Ping the FTP server using *ftp.university.example*.
- Use *nslookup ftp.university.example*, where supported.
- Compare IP-based access and hostname-based access.
- Use Simulation Mode to observe the DNS query and response.

DNS commonly uses UDP or TCP port 53; UDP is typical for ordinary queries, while TCP support is also required for DNS operation.


ADMIN-PC
Physical
Config
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Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping ftp.university.example
Pinging 129.0.0.5 with 32 bytes of data:
Reply from 129.0.0.5: bytes=32 time<1ms TTL=128
Reply from 129.0.0.5: bytes=32 time=1ms TTL=128
Reply from 129.0.0.5: bytes=32 time<1ms TTL=128
Reply from 129.0.0.5: bytes=32 time<1ms TTL=128
Ping statistics for 129.0.0.5:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>


FACULTY_MAIN
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping ftp.university.example
Pinging 129.0.0.5 with 32 bytes of data:
Reply from 129.0.0.5: bytes=32 time<1ms TTL=128
Reply from 129.0.0.5: bytes=32 time=1ms TTL=128
Reply from 129.0.0.5: bytes=32 time<1ms TTL=128
Reply from 129.0.0.5: bytes=32 time<1ms TTL=128
Ping statistics for 129.0.0.5:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>

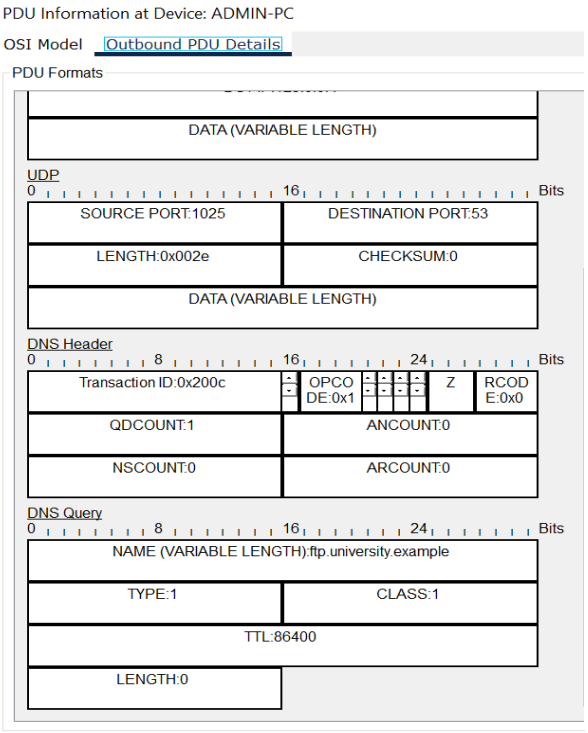

FACULTY REMOTE
Physical
Config
Desktop
Programming
Attributes
Command Prompt
C:\>ping 129.0.0.4
Pinging 129.0.0.4 with 32 bytes of data:
Reply from 129.0.0.4: bytes=32 time=24ms TTL=126
Reply from 129.0.0.4: bytes=32 time=2ms TTL=126
Reply from 129.0.0.4: bytes=32 time=31ms TTL=126
Reply from 129.0.0.4: bytes=32 time=25ms TTL=126
Ping statistics for 129.0.0.4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 31ms, Average = 20ms
C:\>tracert 129.0.0.5
Tracing route to 129.0.0.5 over a maximum of 30 hops:
1 0 ms 0 ms 0 ms 131.0.0.1
2 1 ms 2 ms 3 ms 130.0.0.1
3 0 ms 12 ms 1 ms 129.0.0.5
Trace complete.
C:\>ping ftp.university.example
Pinging 129.0.0.5 with 32 bytes of data:
Reply from 129.0.0.5: bytes=32 time=3ms TTL=126
Reply from 129.0.0.5: bytes=32 time=11ms TTL=126
Reply from 129.0.0.5: bytes=32 time=10ms TTL=126
Reply from 129.0.0.5: bytes=32 time=10ms TTL=126
Ping statistics for 129.0.0.5:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 3ms, Maximum = 11ms, Average = 8ms



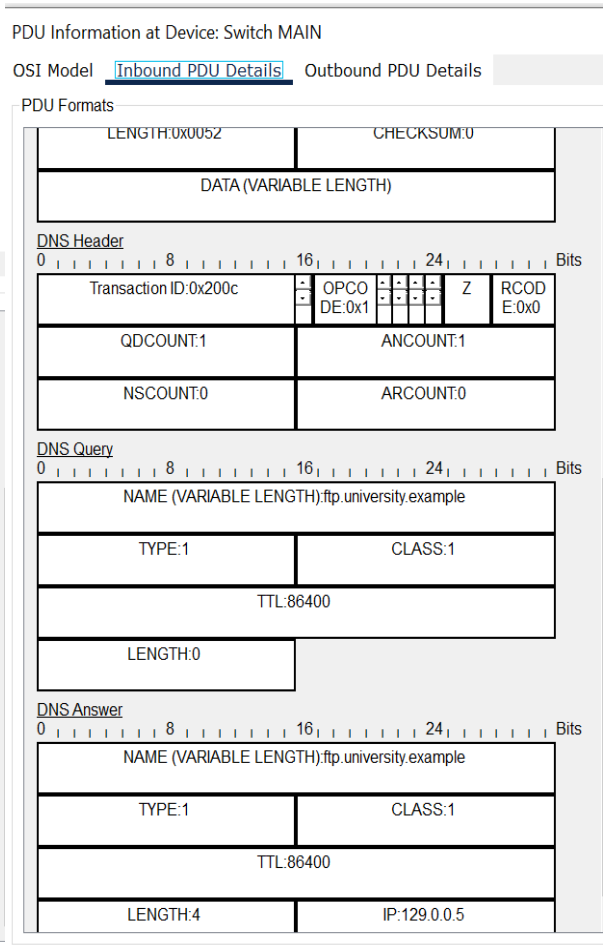
DNS Observation Table:

Client	Query name	DNS-server IP	Returned address	Successful?
Admin-PC	ftp.university.example	129.0.0.4	129.0.0.5	Yes
Faculty-Remote	ftp.university.example	129.0.0.4	129.0.0.5	Yes
Student-PC	ftp.university.example	129.0.0.4	129.0.0.5	Yes
Faculty-Main	ftp.university.example	129.0.0.4	129.0.0.5	yes

DNS Query:



DNS response:



DNS Packet Analysis:

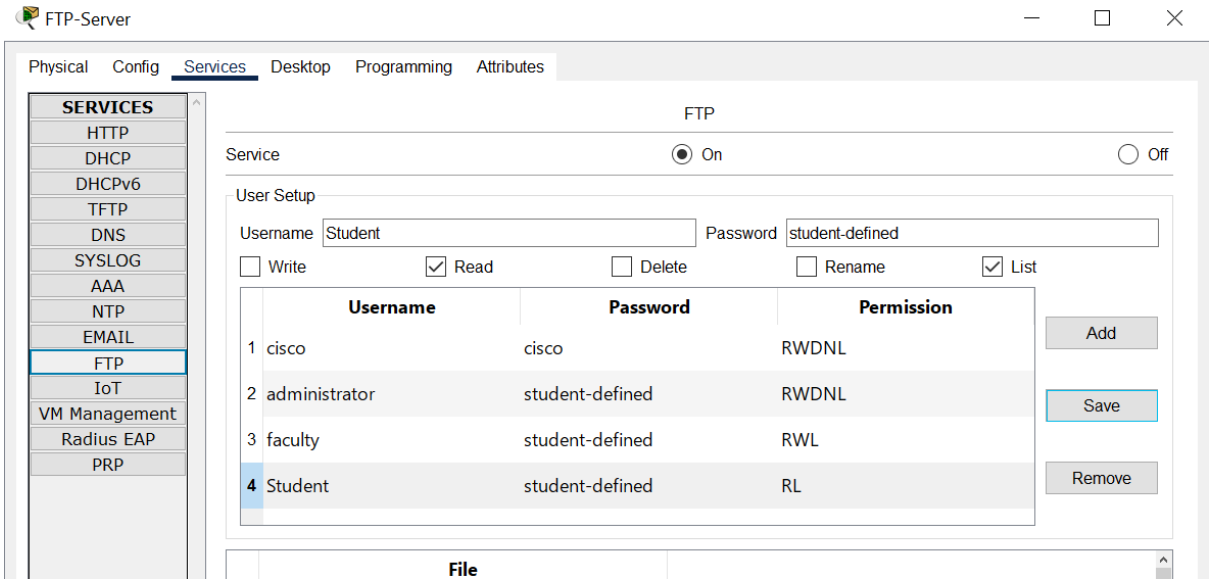
Field	DNS Query	DNS Response
Source IP	129.0.0.2	129.0.0.4
Destination IP	129.0.0.4	129.0.0.2
Source port	1025	53
Destination port	53	1025
Queried domain name	ftp.university.example	ftp.university.example
Returned IP address	N.A.	129.0.0.5
Transport protocol	UDP	UDP

Part C: FTP Server Configuration

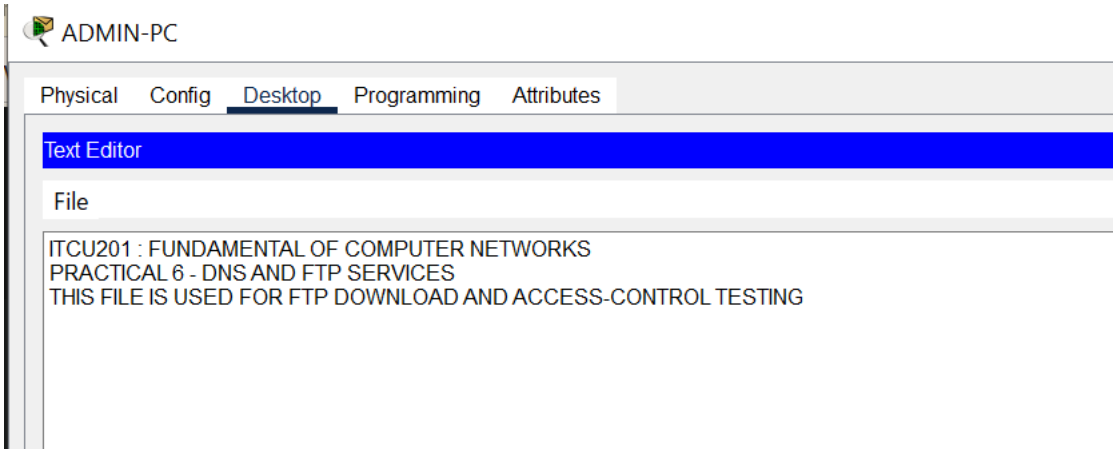
On the FTP server:s

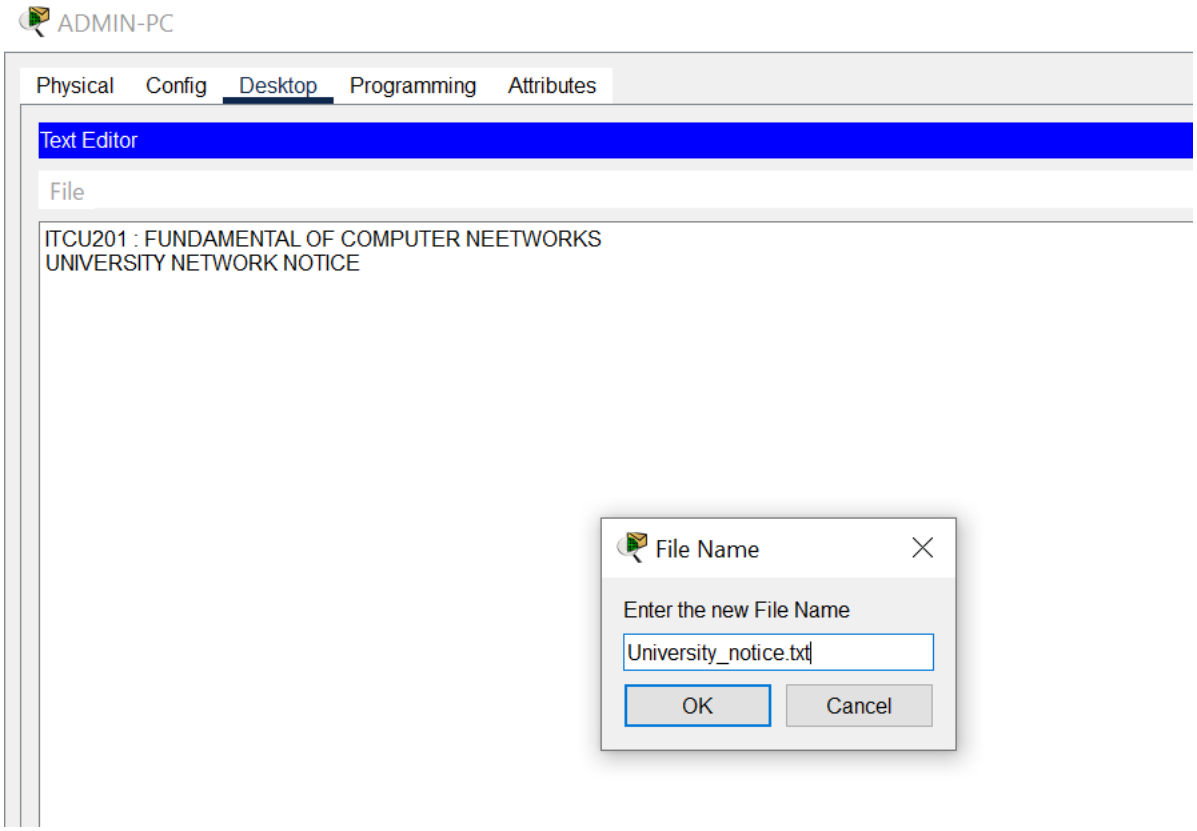
- Open Services → FTP.
- Enable the FTP service.
- Add the required files to the server.
- Create the following user accounts.

Username	Role	Password	List	Read	Write	Delete	Rename
ftp admin	Administrator	Student-defined	Yes	Yes	Yes	Yes	Yes
faculty	Faculty	Student-defined	Yes	Yes	Yes	No	No
student	Student	Student-defined	Yes	Yes	No	No	No



- Passwords should not contain the username or student ID.
- Create at least two sample files:
  - Network_Lab_Manual.txt
  - University_Notice.txt





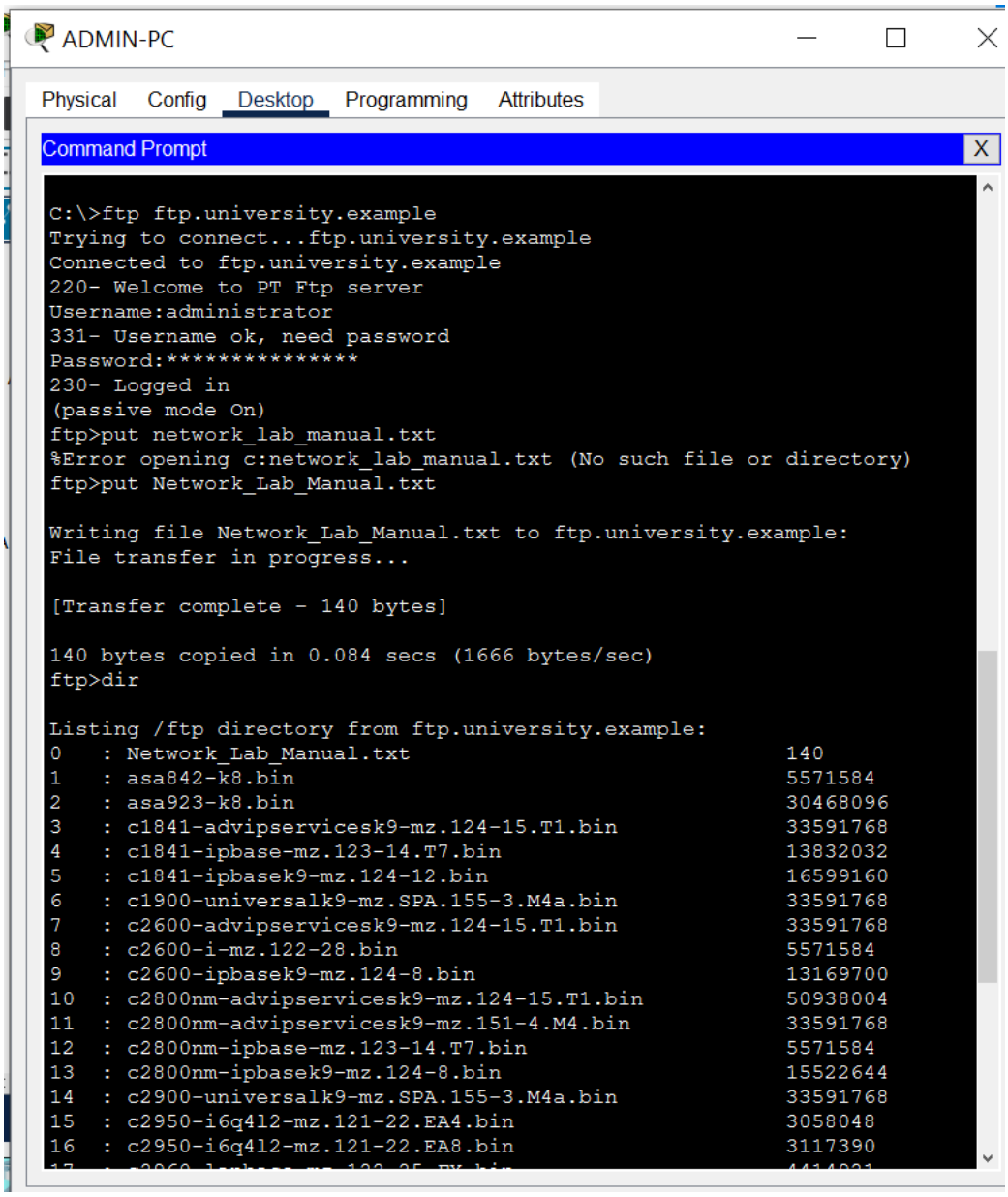
Part D: FTP Access and Permission Testing

Access the FTP server first using its IP address and then using:

ftp.university.example

- Depending on the Packet Tracer version, use the available FTP client or the PC command prompt.
- Typical FTP commands include: ftp ftp.university.example, dir, get <filename>, put <filename>, delete <filename>, quit
- FTP uses a persistent control connection and a separate data connection for operations such as file transfer and directory listing

Test 1: Administrator Account



Perform and insert screenshots:

- Login

```
C:\>ftp ftp.university.example
Trying to connect...ftp.university.example
Connected to ftp.university.example
220- Welcome to PT Ftp server
Username:administrator
331- Username ok, need password
Password:***************
230- Logged in
(passive mode On)
```

- List files

```
140 bytes copied in 0.084 secs (1666 bytes/sec)
ftp>dir

Listing /ftp directory from ftp.university.example:
0    : Network_Lab_Manual.txt                          140
1    : asa842-k8.bin                                   5571584
2    : asa923-k8.bin                                   30468096
3    : c1841-advipservicesk9-mz.124-15.T1.bin          33591768
4    : c1841-ipbase-mz.123-14.T7.bin                   13832032
5    : c1841-ipbasek9-mz.124-12.bin                    16599160
6    : c1900-universalk9-mz.SPA.155-3.M4a.bin          33591768
```

- Download a file

```
ftp>get Network_Lab_Manual.txt

Reading file Network_Lab_Manual.txt from ftp.university.example:
File transfer in progress...

[Transfer complete - 140 bytes]

140 bytes copied in 0 secs
ftp>
```

- Upload a file

```
ftp>put admin_update.txt

Writing file admin_update.txt to ftp.university.example:
File transfer in progress...

[Transfer complete - 12 bytes]

12 bytes copied in 0.088 secs (136 bytes/sec)
ftp>
```

- Rename a file

```
ftp>rename admin_update.txt admin_update_rename.txt

Renaming admin_update.txt
ftp>
[OK Renamed file successfully from admin_update.txt to admin_update_rename.txt]
ftp>dir

Listing /ftp directory from ftp.university.example:
0    : Network_Lab_Manual.txt                          140
1    : admin_update_rename.txt                         12
```

- Delete a file

```
ftp>delete admin_update_rename.txt

Deleting file admin_update_rename.txt from ftp.university.example: ftp>
[Deleted file admin_update_rename.txt successfully ]
ftp>dir

Listing /ftp directory from ftp.university.example:
0    : Network_Lab_Manual.txt                          140
1    : asa842-k8.bin                                   5571584
2    : asa923-k8.bin                                   30468096
```

- Closing an active FTP session

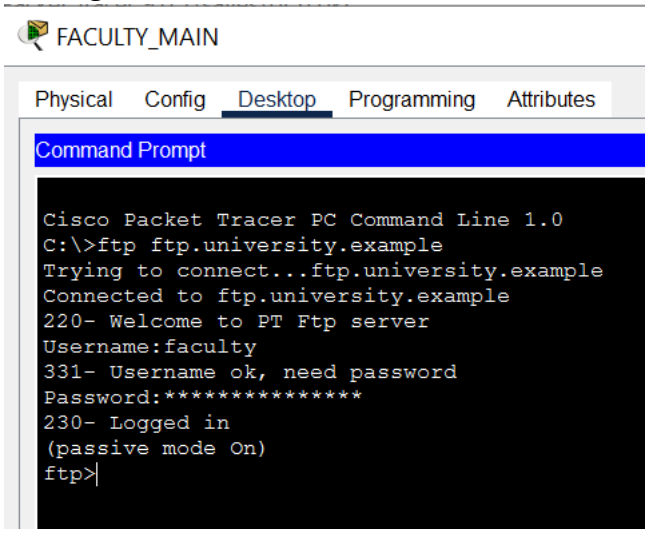
```
ftp>quit

221- Service closing control connection.
C:\>
```

Test 2: Faculty Account

Perform and insert screenshots:

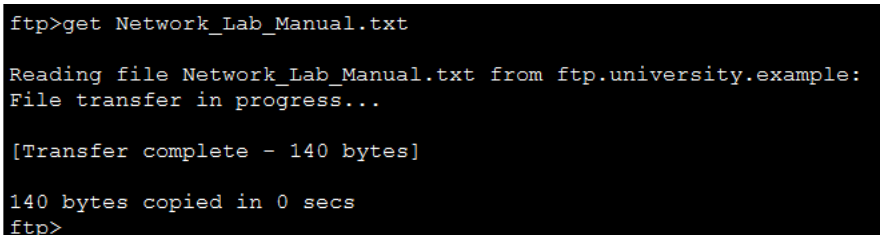
- Login



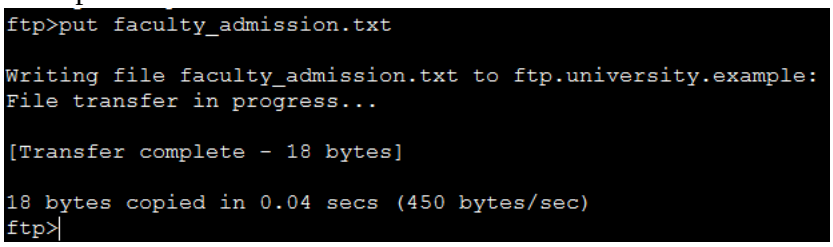
- List files



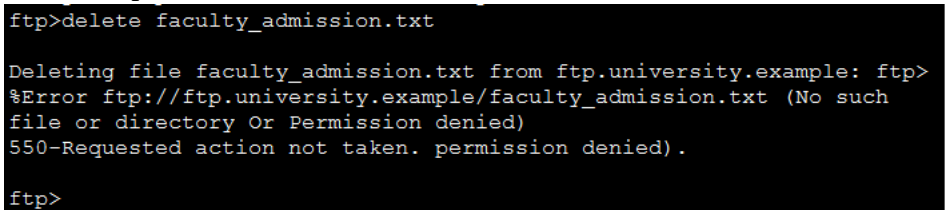
- Download a file



- Upload a file



- Attempt to delete a file



- The deletion attempt should be denied if the permissions are configured correctly.

Test 3: Student Account

Perform and insert screenshots:

- Login

```
C:\>ftp ftp.university.example
Trying to connect...ftp.university.example
Connected to ftp.university.example
220- Welcome to PT Ftp server
Username:student
331- Username ok, need password
Password:***************
230- Logged in
(passive mode On)
ftp>
```

- List files

```
ftp>dir

Listing /ftp directory from ftp.university.example:
0    : Network_Lab_Manual.txt                              140
1    : asa842-k8.bin                                       5571584
2    : asa923-k8.bin                                       30468096
3    : c1841-advipservicesk9-mz.124-15.T1.bin              33591768
4    : c1841-ipbase-mz.123-14.T7.bin                       13832032
```

- Download a file

```
ftp>get Network_Lab_Manual.txt

Reading file Network_Lab_Manual.txt from ftp.university.example:
File transfer in progress...

[Transfer complete - 140 bytes]

140 bytes copied in 0.03 secs (4666 bytes/sec)
ftp>
```

- Attempt to upload a file

```
ftp>delete faculty_admission.txt

Deleting file faculty_admission.txt from ftp.university.example: ftp>
%Error ftp://ftp.university.example/faculty_admission.txt (No such
file or directory Or Permission denied)
550-Requested action not taken. permission denied).

ftp>
```

- Attempt to delete a file

```
ftp>delete student_id.txt

Deleting file student_id.txt from ftp.university.example:
ftp>
%Error ftp://ftp.university.example/student_id.txt (No such file or
directory Or Permission denied)
550-Requested action not taken. permission denied).

ftp>
```

- Upload and deletion attempts should be denied.

FTP Operation Table:

User	Login	List	Download	Upload	Delete	Expected result obtained?
Administrator	Allowed	Allowed	Allowed	Allowed	Allowed	Yes
Faculty	Allowed	Allowed	Allowed	Allowed	Denied	Yes
Student	Allowed	Allowed	Allowed	Denied	Denied	yes

Access-Control Analysis:

Attempted operation	Account used	Allowed/Denied	Permission responsible	Evidence
File download	Student	Allowed	Read	File downloaded successfully
File upload	Student	Denied	Write	Access denied
File upload	Faculty	Allowed	Write	File uploaded
File deletion	Faculty	Denied	Delete	Access denied
File deletion	Administrator	Allowed	Delete	File deleted

- **ftp ftp.university.example**: Connects to the remote FTP server to start a session.
- **dir** : Displays a list of all files and folders stored on the FTP server.
- **get Network_Lab_Manual.txt**: Downloads a file from the remote FTP server onto your local machine.
- **put Admin_Update.txt**: Uploads a file from your local machine to the remote FTP server.
- **rename Admin_Update.txt Admin_Update_Renamed.txt**: Changes the name of a file stored on the FTP server.
- **delete Admin_Update_Renamed.txt**: Permanently removes a specified file from the FTP server.
- **quit**: Terminates the FTP session and returns to the normal command prompt.

Part E: Packet Analysis Using Simulation Mode

Use Packet Tracer Simulation Mode and select relevant event filters such as:

- DNS
- TCP
- FTP

Perform FTP access from the Remote Campus using the hostname.

Observe the following sequence:

- DNS query generated by the client.
- DNS response containing the FTP-server IP address.

- TCP connection establishment with the FTP server.
- FTP authentication request and response.
- FTP command exchange.
- Separate activity for file listing or file transfer.
- TCP connection termination.

Integrated Packet-Flow Table:

Event	Source device	Destination device	Protocol	Source port	Destination port	Purpose
Name-resolution request	131.0.0.3	129.0.0.4	DNS	1025	53	Resolve FTP hostname
Name-resolution response	129.0.0.4	131.0.0.3	DNS	53	1025	Return FTP-server IP
Connection initiation	131.0.0.3	129.0.0.5	TCP	1025	21	Establish FTP control connection
FTP login	129.0.0.5	131.0.0.3	FTP	21	1025	Authenticate user
Directory listing	131.0.0.3	129.0.0.5	FTP	1027	1030	Display server files
File download/upload	131.0.0.3	129.0.0.5	FTP/TCP	1028	1032	Transfer file data

This table outlines the end-to-end communication sequence, starting with DNS name resolution on port 53 before initiating the FTP control connection on port 21. It demonstrates how dynamic ephemeral client ports pair with well-known server ports to facilitate authentication, directory listing, and data transfers across the network. Overall, it highlights the transition from initial UDP-based DNS lookups to persistent TCP-based application data sessions.

Address Observation:

For one remote FTP session, record:

Network segment	Source IP	Destination IP	Source MAC	Destination MAC
Remote client LAN	129.0.0.5	131.0.0.3	000A.41EE.3ED4	00E0.8F91.0C1A
WAN segment	129.0.0.5	131.0.0.3	N.A. if serial	N.A. if serial
Main-campus server LAN	129.0.0.5	131.0.0.3	000A.41EE.3ED4	00E0.8F91.0C1A

This table demonstrates how Layer 3 IP addresses remain constant (126.0.0.3 to 124.0.0.3) across all network segments during end-to-end communication. Conversely, Layer 2 MAC addresses are local to their specific LAN segments and are omitted entirely on point-to-point WAN serial links. This illustrates fundamental OSI model behaviour, where logical IP routing spans the whole path while physical MAC addressing changes at every hop.

Part F: Troubleshooting Activities

Introduce one fault at a time.

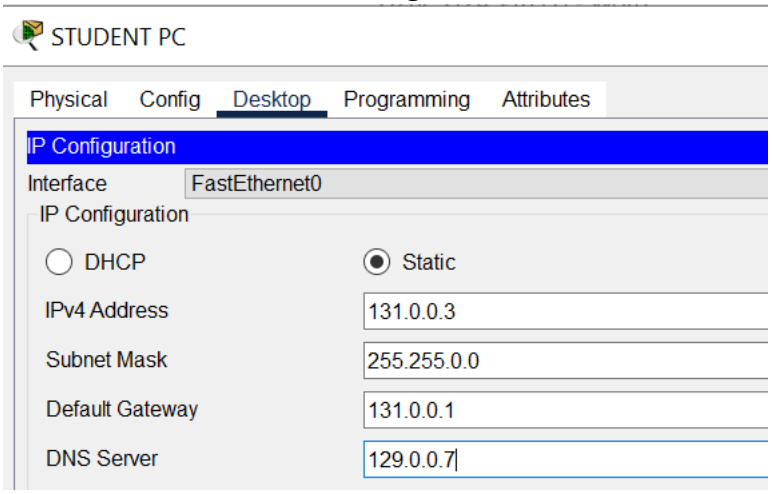
Fault 1: Incorrect DNS-Server Address

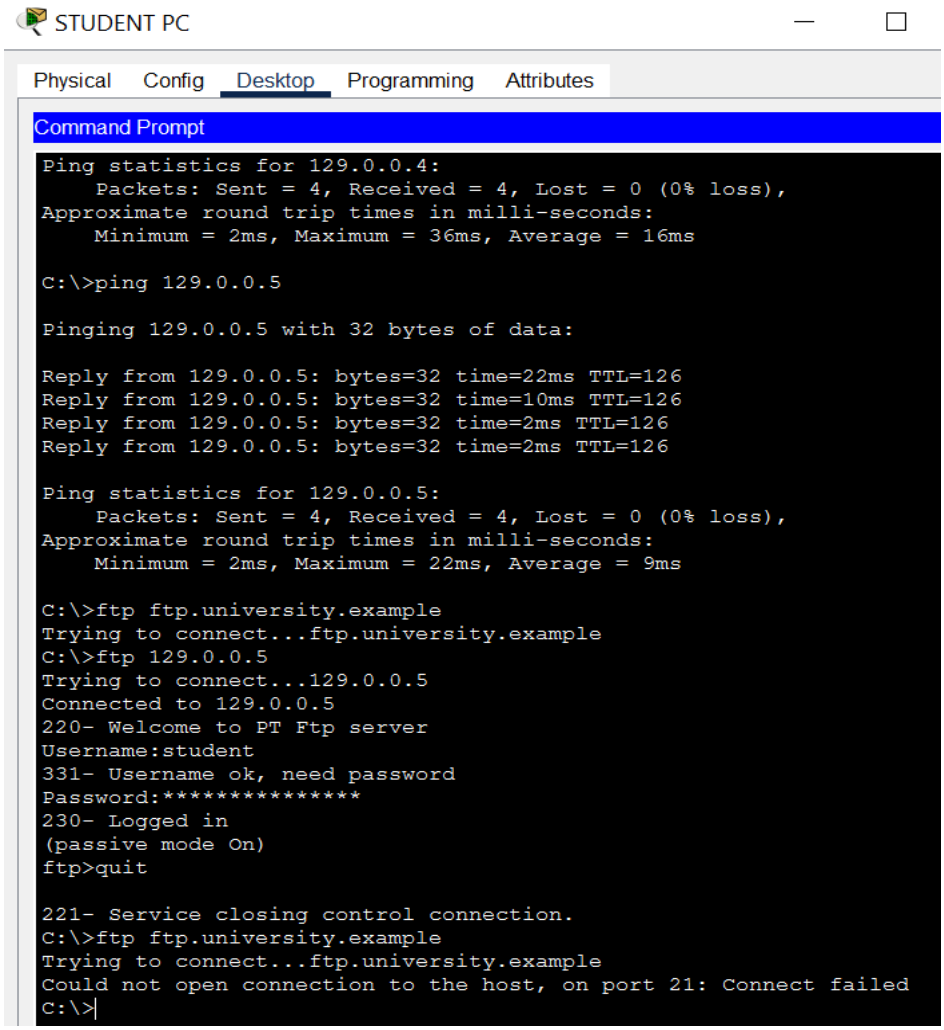
Change the DNS-server address on the Student-PC.

Verify:

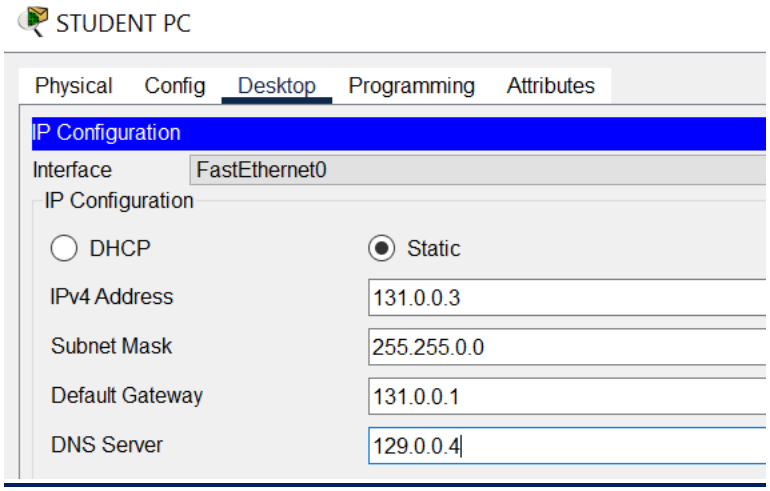
- Ping to FTP-server IP
- Ping to *ftp.university.example*
- FTP access using IP
- FTP access using hostname

Insert screenshots and write diagnosis:





Connection:



The Student-PC was configured with an incorrect DNS server address. However, the FTP server remained reachable using its IP address, which confirmed that basic network connectivity and routing were working. The hostname ftp.university.example could not be resolved because the client was not communicating with the correct DNS server. The problem was corrected by restoring the DNS server address to 129.0.0.2. After correction, hostname resolution and FTP access using the domain name worked successfully.

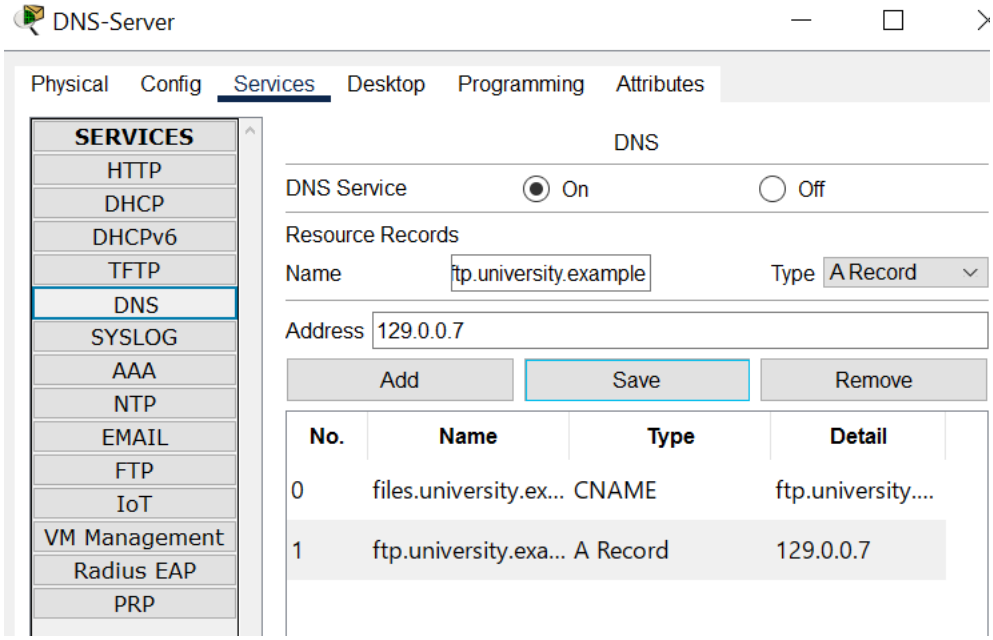
Fault 2: Missing or Incorrect DNS Record:

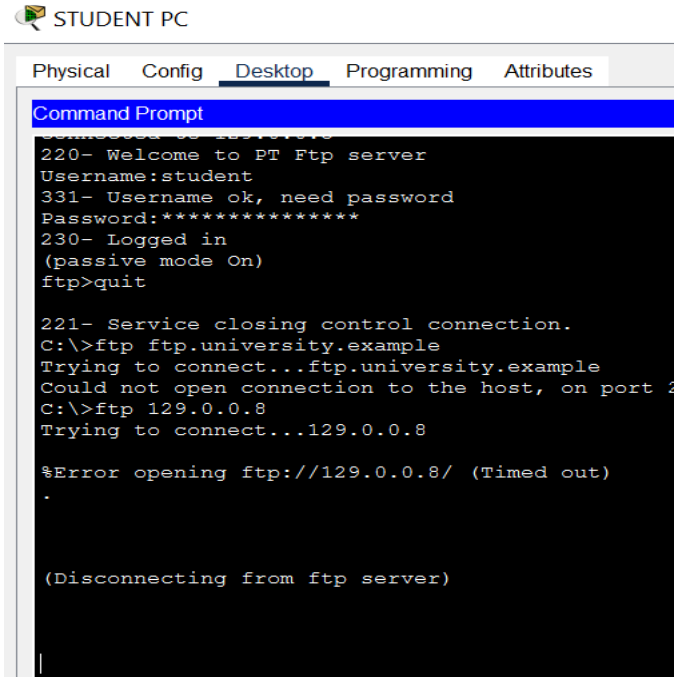
Delete or modify the A record for *ftp.university.example*.

Identify:

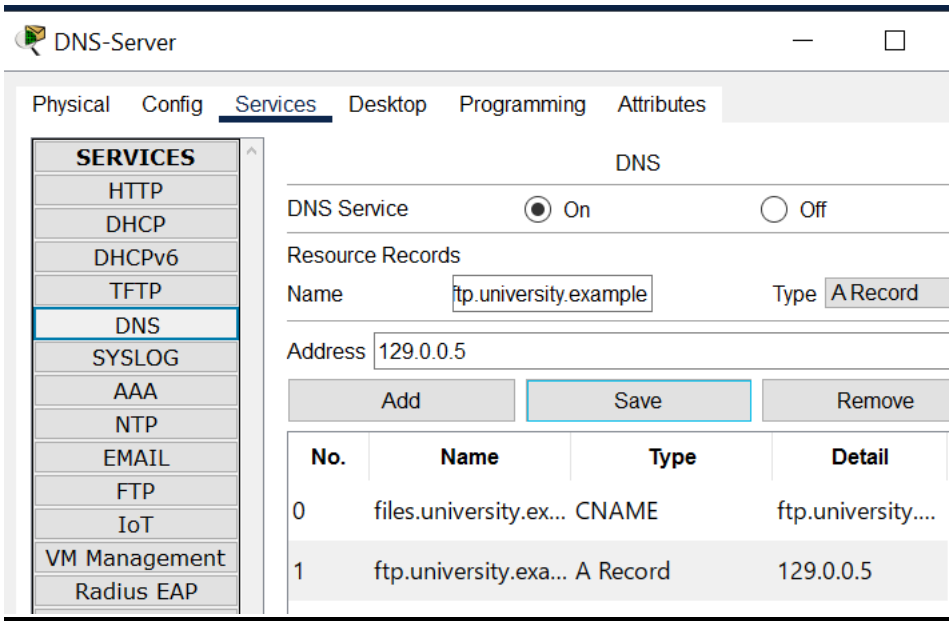
- DNS query result
- Incorrect or missing returned IP address
- Difference between IP-based and name-based FTP access

Insert screenshots and write diagnosis:





Connections:



The A record for ftp.university.example was missing or incorrect. The FTP server could still be accessed using its IPv4 address, proving that IP connectivity and the FTP service were available. However, hostname-based access failed because DNS could not return the correct FTP server address. The correct A record is ftp.university.example → 124.0.0.3. After restoring the DNS record, hostname resolution and FTP access using the domain name worked successfully.

Fault 3: FTP Service Disabled

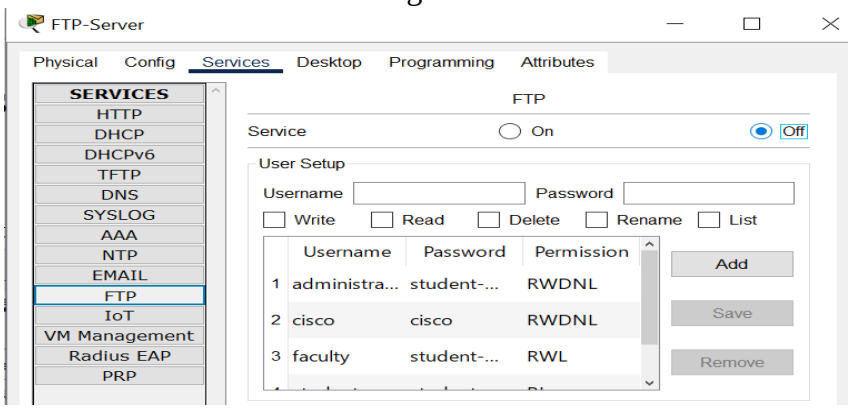
Disable the FTP service while keeping DNS operational.

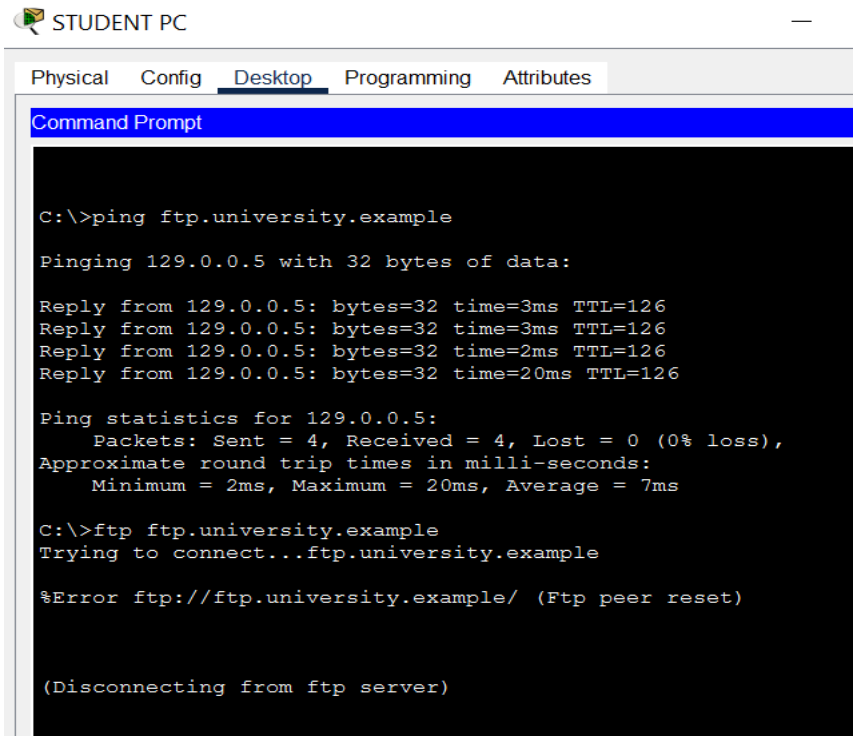
Insert screenshots and write diagnosis:

Fault 4: Incorrect User Credentials

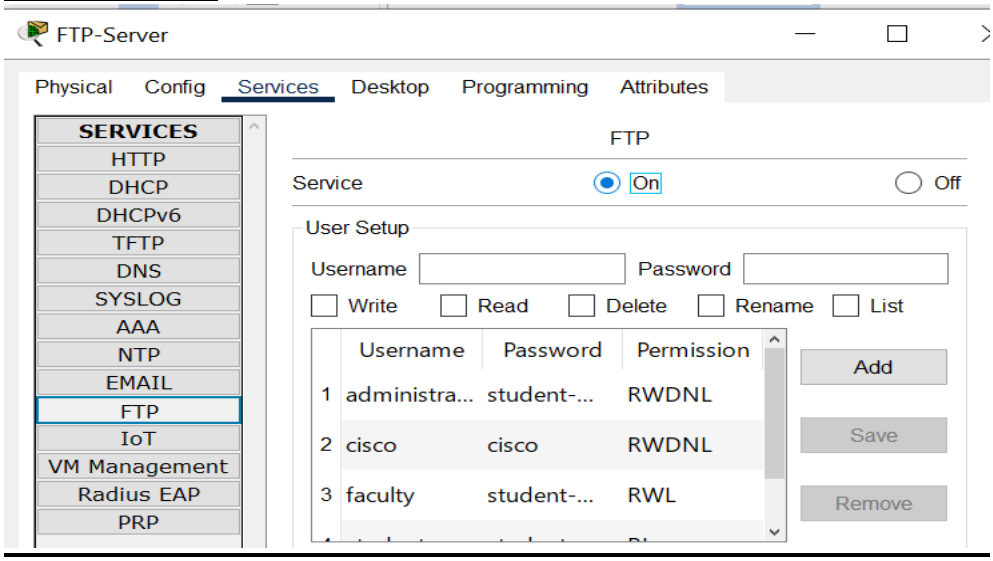
Enter an incorrect username or password.

Insert screenshots and write diagnosis:




STUDENT PC
Physical
Config
Desktop
Programming
Attributes
Command Prompt
C:\>ping ftp.university.example
Pinging 129.0.0.5 with 32 bytes of data:
Reply from 129.0.0.5: bytes=32 time=3ms TTL=126
Reply from 129.0.0.5: bytes=32 time=3ms TTL=126
Reply from 129.0.0.5: bytes=32 time=2ms TTL=126
Reply from 129.0.0.5: bytes=32 time=20ms TTL=126
Ping statistics for 129.0.0.5:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 20ms, Average = 7ms
C:\>ftp ftp.university.example
Trying to connect...ftp.university.example
%Error ftp://ftp.university.example/ (Ftp peer reset)
(Disconnecting from ftp server)

Correction:

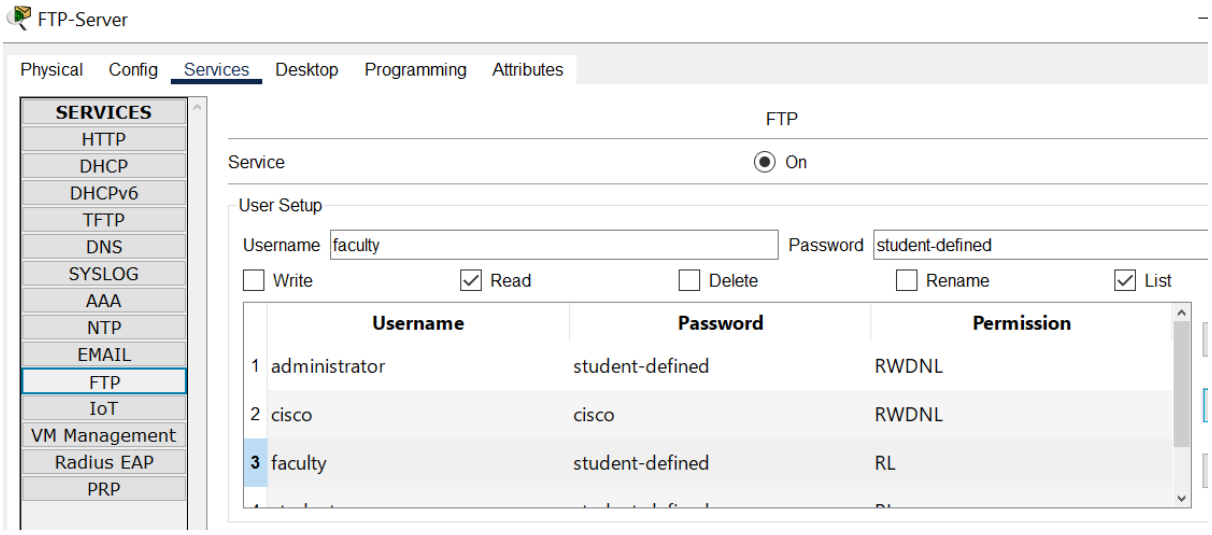

FTP-Server
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
FTP
Service
On
Off
User Setup
Username
Password
Write
Read
Delete
Rename
List
Username
Password
Permission
1 administra... student-... RWDNL
2 cisco cisco RWDNL
3 faculty student-... RWL
Add
Save
Remove

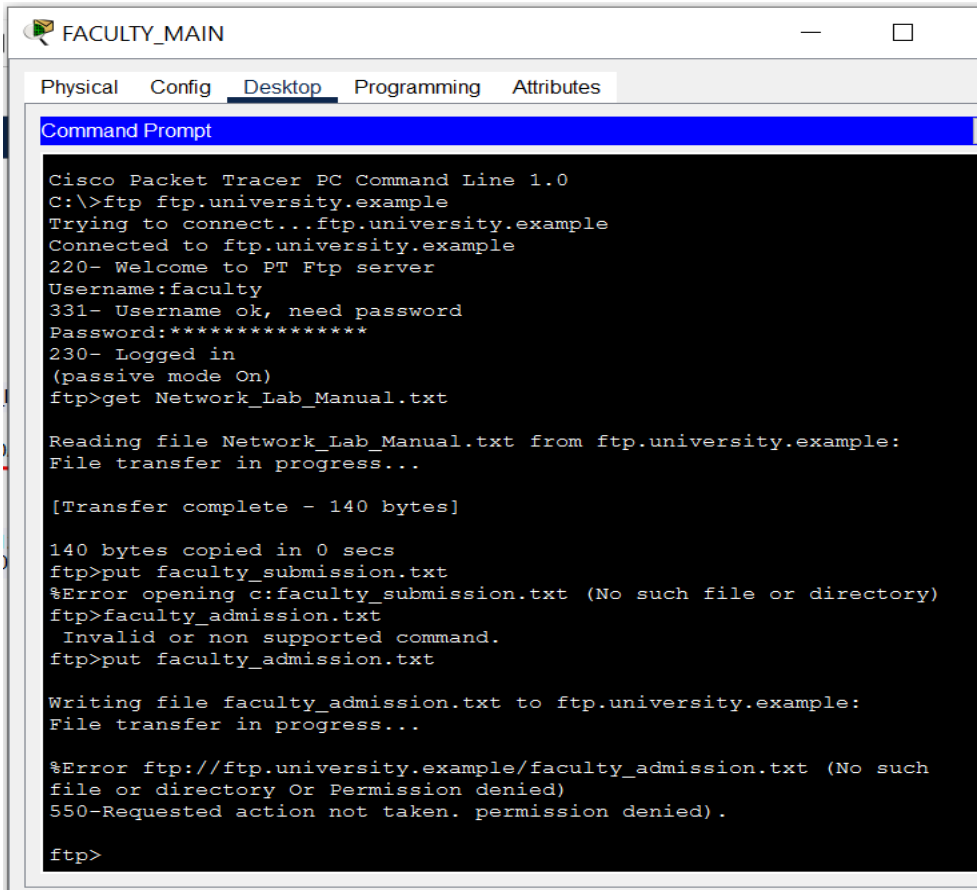
Fault 5: Incorrect Permissions

Remove write permission from the faculty account or accidentally grant write permission to the student account.

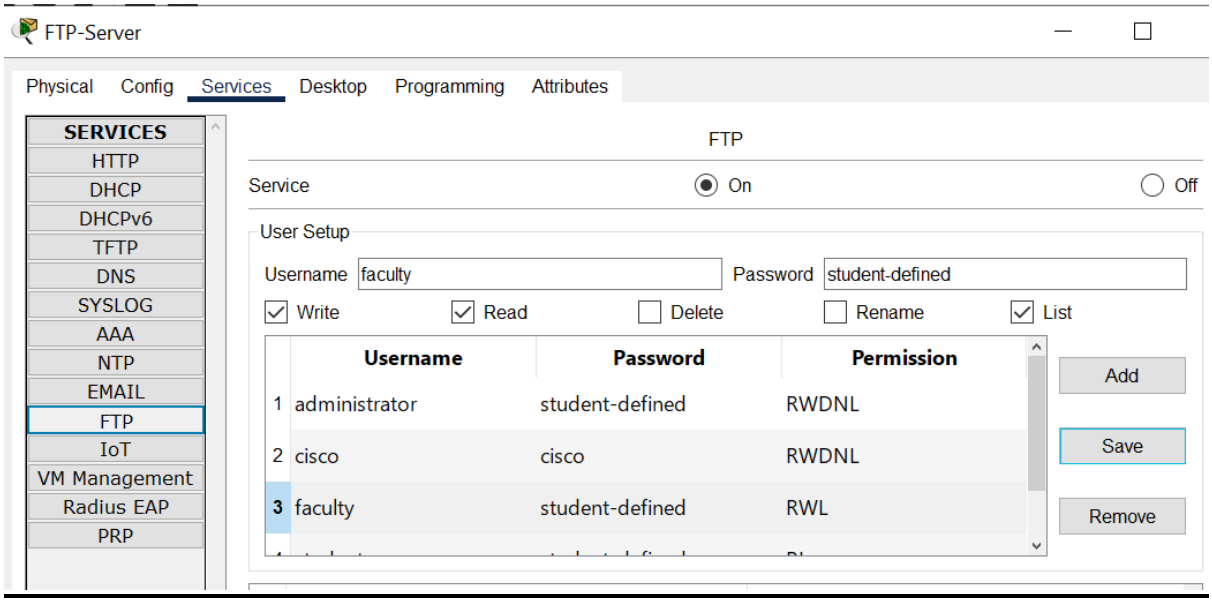
Identify the misconfiguration and restore the required role-based permissions.

Insert screenshots and write diagnosis:





Correction:



The FTP account permissions were intentionally modified to create an access-control fault. The faculty account was tested against its required permissions. When an operation such as deletion was attempted without the required Delete permission, the server denied the operation. This confirmed that FTP permissions were being enforced according to the configured role. The correct role-based permissions were then restored: Administrator with full permissions, Faculty with list/read/write permissions, and Student with list/read permissions only.

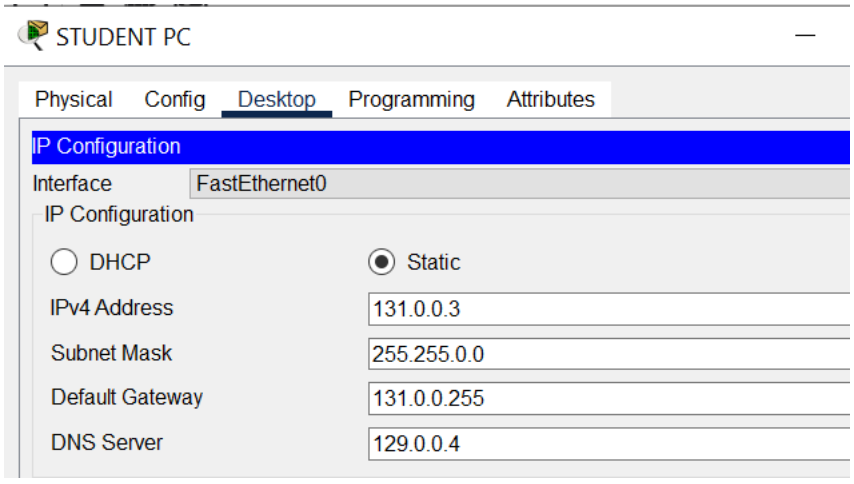
Fault 6: Routing or Gateway Failure

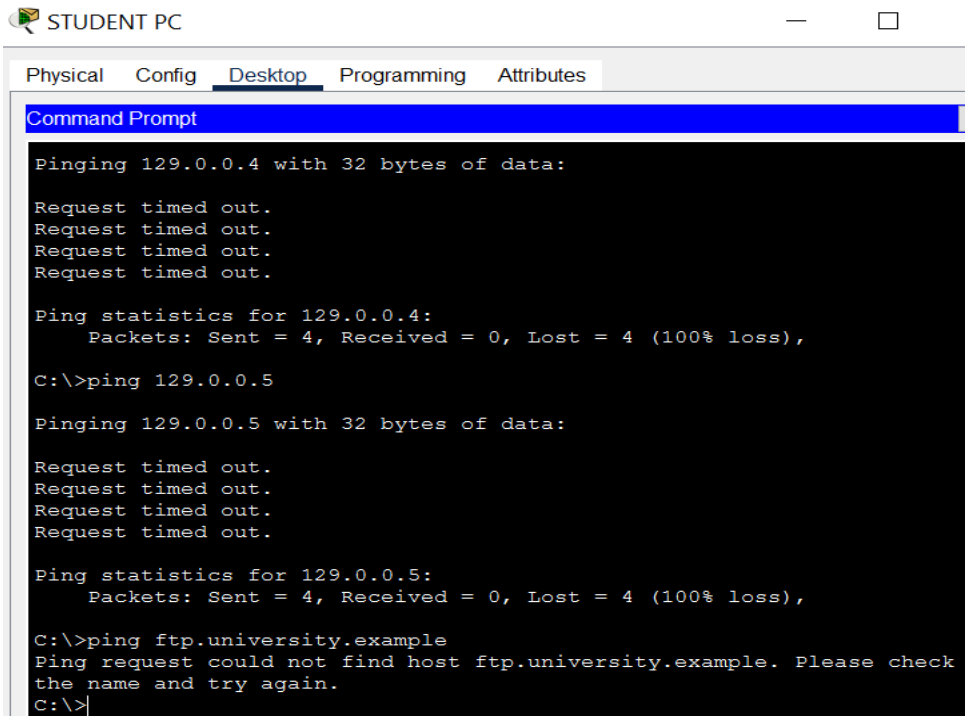
Configure an incorrect default gateway on one remote client.

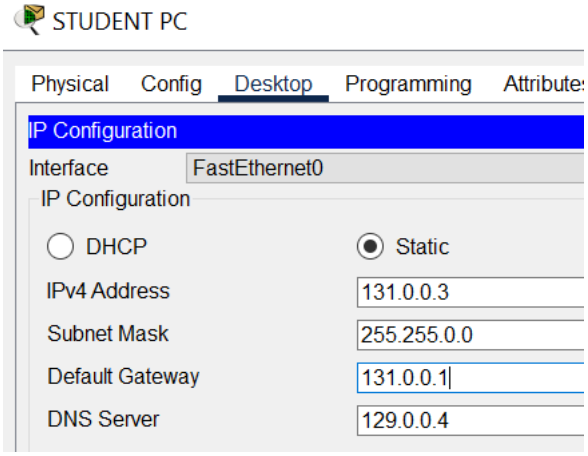
Use: *ipconfig, ping, tracert*

to determine whether the problem is related to DNS, FTP or basic network reachability.

Insert screenshots and write diagnosis:







An incorrect default gateway was configured on the remote client. The incorrect gateway prevented the client from reaching networks outside its local LAN. The ipconfig command was used to verify the configuration, while ping and tracert were used to identify the connectivity problem. The fault was caused by the incorrect default gateway. Restoring the gateway to 126.0.0.1 restored communication with the Main Campus FTP server.

Troubleshooting Record:

Fault	Observed symptom	Diagnostic test	Root cause	Corrective action	Verification
Incorrect DNS address	IP access works; hostname fails	ipconfig, ping IP/name	Wrong client DNS IP	Configure 129.0.0.4	Hostname works
Missing DNS record	P FTP works; hostname fails	DNS Simulation / nslookup	Missing/wrong A record	Restore A record	FTP by name works
FTP disabled	Ping/name resolution works; FTP fails	Simulation TCP/FTP	FTP service Off	Enable FTP	Login succeeds
Invalid credentials	Connection established ; login rejected	Attempt login	Wrong username/password	Use valid credentials	Login succeeds
Incorrect permissions	Login works; operation incorrect	put/delete/get	Wrong permission checkbox	Restore role permissions	Expected operation succeeds/fails
Wrong gateway	Remote IP services unreachable	ipconfig, ping, tracert	Incorrect default gateway	Restore 131.0.0.1	Remote ping succeeds

Part G: FTP Security Analysis

Traditional FTP is suitable for demonstrating control connections, data transfer, authentication and file permissions, but it should not be treated as a secure production file-transfer mechanism.

FTPS adds TLS-based security to FTP, while SFTP is associated with the SSH protocol family and is a different protocol rather than "FTP over SSH."

Complete the following comparison:

Feature	FTP	FTPS	SFTP
Underlying technology	FTP over TCP	FTP + TLS	SSH File Transfer Protocol
Encryption	No	Yes	Yes
Credential protection	No	Yes when TLS is used	Yes
Multiple connections	Separate control/data connections	FTP-style control/data connections	Normally one SSH connection
Suitable for confidential data	No	Yes	Yes
Packet Tracer implementation	Supported	Generally conceptual/not provided by basic ServerPT FTP service	Generally conceptual/not provided as standard endhost service

Standard FTP lacks encryption and exposes credentials in cleartext, making it suitable only for basic simulation environments like Packet Tracer. FTPS enhances standard FTP by layering TLS encryption over separate control and data channels, while SFTP completely replaces FTP with a secure, single-connection protocol running over SSH. Consequently, both FTPS and SFTP are recommended for handling confidential data, whereas traditional FTP is unsafe for production environments.

Security Decision Scenarios:

Requirement	Recommended protocol	Justification
Publicly downloadable non-sensitive files in a controlled lab	FTP	Encryption is not essential and FTP demonstrates basic file transfer
Confidential university records	SFTP	Strong encrypted SSH-based file transfer
Automated secure server-to-server transfer	SFTP	Encryption plus strong automation/key-based authentication support
Legacy system requiring FTP compatibility with encryption	FTPS	Retains FTP-oriented compatibility while adding TLS encryption.

The protocol selection directly aligns security capabilities with specific operational requirements. Unencrypted FTP is ideal for public, non-sensitive lab files and legacy environments that upgrade to TLS (FTPS) for compatibility. Conversely, SFTP is selected for confidential records and automated server transfers due to its robust SSH-based encryption, single-port efficiency, and support for key-based authentication.

Conclusion:

This practical successfully demonstrated the integration of DNS and FTP services across a routed LAN-WAN environment. DNS was configured to resolve ftp.university.example to the FTP server, and connectivity was verified from both Main and Remote Campus clients. Role-based FTP accounts were configured for Administrator, Faculty and Student users, and their permissions were tested through login, listing, download, upload and deletion operations. Packet Tracer Simulation Mode was used to observe DNS queries, DNS responses, TCP connection establishment, FTP authentication and file-transfer communication. Different DNS, FTP, authentication, permission and gateway faults were also introduced and diagnosed. Finally, FTP, FTPS and SFTP were compared based on encryption and security requirements. The practical demonstrated how DNS, routing, FTP services and access-control permissions work together in an interconnected network.

Post-Lab Questions:

1. A remote-campus student can successfully ping the FTP server's IP address, confirming basic network reachability, but cannot establish an FTP session. Develop an application-layer troubleshooting procedure. Verify whether the FTP service is enabled, whether the required user account and credentials exist, whether the client is using the correct server address and whether TCP port 21 is reachable. If the client uses a hostname instead of an IP address, also verify the DNS record and client DNS-server configuration. Explain how successful FTP access should be confirmed through login, directory listing and an authorized file operation.

   First, verify basic network connectivity by pinging the FTP server's IP address. Since the IP address is reachable, the problem is likely at the application layer. Next, verify that the FTP service is enabled on the server and that TCP port 21 is available for the FTP control connection. Check that the required FTP user account exists and verify the username and password. Confirm that the client is using the correct FTP server address. If a hostname is being used, verify the client's DNS server address and confirm that the DNS A record for ftp.university.example points to the correct FTP server IP address. After correcting the problem, confirm FTP access by successfully logging in, listing the directory and performing an authorized file operation such as downloading or uploading a file.

2. A faculty user can access the FTP server using its IPv4 address but cannot access it using *ftp.university.example*. Identify the DNS-related causes, including a missing or incorrect A record, an incorrect DNS-server address on the client, a spelling error in the hostname or an inactive DNS service. State the required DNS resource record and use hostname resolution, *nslookup* where supported, and Packet Tracer Simulation Mode to verify the DNS query and response. After correcting DNS, establish an FTP session using the hostname to confirm complete service access.

   If FTP works using the IPv4 address but not using `ftp.university.example`, the problem is related to DNS rather than basic network connectivity. Possible causes include a missing or incorrect A record, an incorrect DNS server address configured on the client, a spelling mistake in the hostname, or a disabled DNS service. The required DNS record is:

   ```
   ftp.university.example → 124.0.0.3
   ```

   The problem can be verified using `ping ftp.university.example`, `nslookup ftp.university.example` where supported, and Packet Tracer Simulation Mode to observe the DNS query and response. After correcting the DNS configuration, FTP should be tested again using the hostname.

3. A university requires an administrator to list, upload, download and delete files; faculty users to list, upload and download files but not delete them; and students to list and download files only. Prepare a permission matrix for the three user roles and configure the corresponding FTP accounts. Explain how excessive permissions could result in

unauthorized modification or deletion, and how insufficient permissions could prevent legitimate academic work.

The Administrator account has full file-management permissions. The Faculty account can list, read/download and upload files but cannot delete or rename them. The Student account can list and download files but cannot upload, modify, rename or delete files. Excessive permissions could allow users to modify or delete files without authorization. Insufficient permissions could prevent legitimate academic activities such as uploading required files. Therefore, permissions should be assigned according to the user's role and minimum required access.

4. A student successfully authenticates, lists available files and downloads a file, but an upload attempt is denied. A faculty user can list, download and upload files but cannot delete them. Compare these results with the intended permission policy and determine whether access control is working correctly. Describe a systematic test procedure for each role covering valid login, invalid login, directory listing, download, upload, deletion and expected access-denied operations.

   The results match the intended access-control policy. A Student is allowed to authenticate, list files and download files, but upload and deletion operations should be denied. A Faculty user is allowed to list, download and upload files, but deletion should be denied. Therefore, these results indicate that role-based access control is functioning correctly.
   A systematic testing procedure should include:
   1. Test valid login.
   2. Test invalid username/password.
   3. Test directory listing.
   4. Test file download.
   5. Test file upload.
   6. Test file deletion.
   7. Verify that operations expected to be denied are actually denied.
   8. Repeat the tests for Administrator, Faculty and Student accounts.

   The observed results should then be compared with the required permission matrix.

5. A remote user accesses an FTP server using its domain name and downloads a file. Using Packet Tracer Simulation Mode, describe the expected sequence beginning with DNS name resolution and continuing through FTP authentication and file transfer. Identify the DNS query and response, the TCP three-way handshake for the FTP control connection, the purpose of TCP server port 21, login and command exchange, and the separate TCP data connection used for directory listing or file download. Record the data-connection ports actually observed in Simulation Mode and identify whether the implementation represents active or passive FTP behaviour.

   When the remote client accesses the FTP server using ftp.university.example, the first step is DNS name resolution. The client sends a DNS query to the DNS server using UDP port 53. The DNS server responds with the FTP server's IP address, 124.0.0.3.

After DNS resolution, the client establishes the FTP control connection with the FTP server using TCP. The TCP three-way handshake consists of SYN, SYN-ACK and ACK packets. The FTP server uses TCP port 21 for the control connection.

After the TCP connection is established, the client sends the FTP username and password. The server authenticates the user and accepts FTP commands such as directory listing and file download. Directory listing and file transfer use a separate FTP data connection.

In the Simulation Mode analysis, the DNS communication uses port 53 and the FTP control connection uses destination port 21. The exact data-connection source and destination ports should be recorded from the Packet Tracer Simulation Mode packet details. These ports are implementation/session dependent and should not be guessed.

The observed data-connection behaviour should then be identified as active or passive based on the actual FTP packets and commands shown in Simulation Mode.

6. Three departments report different FTP problems:
   - Department A cannot resolve the FTP hostname.
   - Department B can reach the server but authentication fails.
   - Department C can authenticate but cannot upload files.

   Identify the most likely DNS, account or permission fault in each case and state the evidence required to confirm the diagnosis. Then recommend an appropriate protocol from FTP, FTPS and SFTP for each of the following:

   a. Public, non-sensitive file distribution where compatibility is the main concern
   b. Confidential student-record exchange with an existing FTP-based system
   c. Automated secure server-to-server transfer

   Justify each recommendation with respect to encryption, authentication, permission control, firewall compatibility and automation.

   Department A: The most likely problem is a DNS configuration fault. The hostname cannot be resolved, so the DNS service, DNS server address on the client and A record should be checked. Evidence can be obtained using ping with the hostname, nslookup where supported, and Simulation Mode to observe the DNS query and response.

   Department B: The most likely problem is an account or credential fault. Since the server can be reached but authentication fails, the username, password and existence of the FTP account should be checked. A successful network connection followed by an authentication failure provides evidence that the problem occurs at the authentication stage.

   Department C: The most likely problem is an FTP permission fault. If the user can authenticate but cannot upload, the Write permission for that account should be

checked. For example, the Student role has Write permission disabled according to the required policy. The FTP server's permission configuration and the result of a put operation can be used as evidence.

Protocol recommendations:

a. **Public, non-sensitive file distribution where compatibility is the main concern:** FTP can be used in the controlled lab scenario because encryption is not required for non-sensitive data and compatibility is important.

b. **Confidential student-record exchange with an existing FTP-based system:** FTPS is appropriate when FTP compatibility needs to be retained while adding TLS encryption.

c. **Automated secure server-to-server transfer:** SFTP is appropriate because it uses SSH-based encryption and supports secure automated transfers and key-based authentication.